

Boat

→ **Source**

```
import java.util.*;

public class Main{
    static int maxTeams(int[] arr,int[] freq){
        int result =0;
        for(int s =2;s<=2*arr.length;s++){
            int t =0;
            for(int x=1;x<=arr.length;x++){
                int t1=s-x;
                if(t1>arr.length||t1<=0) continue;
                if(x>t1) continue;
                if(x<t1) t+=Math.min(freq[x],freq[s-x]);
                else
                    t+=freq[x]/2;
            }
            result=Math.max(result,t);
        }
        return result;
    }
    public static void main(String[] args){
        Scanner in = new Scanner(System.in);
        int t = in.nextInt();
        for(int i = 0;i<t;i++){
            int n = in.nextInt();
            int[] arr = new int[n];
            int[] freq = new int[n+1];
            for(int j =0;j<n;j++){
                arr[j]=in.nextInt();
                freq[arr[j]]++;
            }

            System.out.println(maxTeams(arr,freq));
        }
    }
}
```

Three parts of array

→ Source

```
import java.util.*;

public class Main{
    public static void main(String[] args){
        Scanner sc = new Scanner(System.in);
        int n = sc.nextInt();
        int[] arr = new int[n];
        for(int i =0;i<n;i++)
            arr[i]=sc.nextInt();
        int l =0;
        int r= n-1;
        long s1= arr[l];
        long s3 = arr[r];
        long max = 0;
        while(l<r){
            if(s1==s3){
                max = s1;
                l++;
                s1+=arr[l];
            }
            else if(s1<s3){
                l++;
                s1+=arr[l];
            }
            else if(s1>s3){
                r--;
                s3+=arr[r];
            }
        }
        System.out.println(max);
    }
}
```
